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PROJECT WORK REPORT ON

“Smart Home Automation-Based Hand Gesture Recognition Using Mediapipe and Raspberry Pi ”

Submitted in the partial fulfillment of the requirement for the award
of degree

BACHELOR OF ENGINEERING IN ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

Certified that the Project work Synopsis entitled **“Smart Home Automation-Based Hand Gesture Recognition Using Feature Fusion and Recurrent Neural Network ”** carried out by, **Bhavan Kumar V (1EW22EC025), Dhanush E(1EW22EC036), Deekshitha KS (1EW23EC402), Harshitha K(1EW23EC407)**, is a bonified student of **ELECTRONICS AND COMMUNICATION** in partial fulfillment for the award of degree of Bachelor of Engineering in **East West Institute of Technology** of the Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections suggestions indicated for Internal Assessment have been incorporated in the synopsis deposited in the department library. This project work synopsis has been approved as it satisfies the academic requirement in respect of project work prescribed for the Bachelor of Engineering Degree.

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Smart Home Automation using Hand Gestures with Mediapipe and Raspberry Pi

Abstract

Smart home automation systems have gained significant popularity in recent years, enhancing comfort, safety, and energy efficiency in modern households. Traditional smart home systems often rely on mobile applications, voice assistants, or remote controls. However, these approaches can sometimes be inconvenient or inaccessible for certain users. In this project, we propose a gesture-based smart home automation system using the Mediapipe library integrated with Raspberry Pi 3B+. A camera captures hand gestures, which are processed in real-time using Python and Mediapipe. Based on the recognized gestures, three electrical appliances are controlled via a 3-channel relay board connected to the Raspberry Pi. This project demonstrates a contactless, low-cost, and user-friendly approach to controlling home appliances, offering accessibility for differently-abled individuals and convenience for general users.

Introduction

Smart homes integrate various technologies to provide automated and efficient control over household appliances. Common smart home solutions include smartphone apps, IoT devices, and voice-activated assistants. However, these solutions either require internet connectivity, specialized hardware, or interaction through touch/voice. Hand gesture recognition provides a more natural, intuitive, and contactless interface. With the advances in computer vision libraries such as Mediapipe, gesture recognition has become computationally feasible on small devices like the Raspberry Pi 3B+. This project aims to create a smart home automation system where hand gestures replace traditional switches or voice commands for operating electrical devices.

In recent years, home automation has emerged as a research topic. Many researchers have started investigating the demand criteria for home automation in different environments. Human-computer interaction (HCI) is considered a more interactive and resourceful method of engaging with different appliances to make the system work. In the conventional approach, different devices like a mouse, keyboard, touch screen, and remote devices are used to fulfill requirements so that users can interact by only using their hands with different home appliances, home healthcare, and home monitoring systems. Usually, changing channels and controlling light on/off switches are more demanding research areas for HCI.

Earlier systems were divided into two approaches for interacting with computers. The first approach.

LITERATURE SURVEY

1. Smart Home Control Using Real-Time Hand Gesture Recognition and AI on Raspberry Pi 5.

In this 2025 study by Thomas Hobbs and Anwar Ali from Swansea University, the authors developed a gesture-controlled smart home system using MediaPipe and AI models deployed on Raspberry Pi 5. The system achieved high accuracy in recognizing hand gestures and controlling devices in real time. A key focus was on low-latency processing through edge computing, making the solution efficient and responsive for practical home automation scenarios.

2. Gesture-Controlled Smart Home System Using MediaPipe and Arduino Integration A GitHub-based project by developer damisotomi in 2024 showcased a practical implementation of hand gesture recognition using MediaPipe and OpenCV. The system was designed to detect gestures and transmit commands via serial communication to an Arduino, which controlled various smart home devices. This open-source project emphasized accessibility and reproducibility, making it a valuable resource for students and hobbyists exploring gesture-based automation.

3. Magic Mirror Interface for Smart Home Automation Using Raspberry Pi reTerminal DM Kasun Thushara's 2024 project, published on Seeed Studio's blog, introduced a "Magic Mirror" concept that combined MediaPipe, OpenCV, and Home Assistant. The system used Raspberry Pi reTerminal DM and Xiao ESP32 to enable gesture-based control of smart devices. Designed with hygiene and accessibility in mind, the project allowed users to interact with their home environment without physical contact, making it especially relevant during health-conscious times.

4. Emerging Trends in Gesture-Based Smart Home Automation with MediaPipe and Raspberry Pi Throughout 2024 and 2025, numerous academic and hackathon projects explored the use of MediaPipe's holistic tracking for multi-hand gesture recognition in smart home systems. These implementations often integrated MQTT protocols and platforms like Home Assistant to enable seamless device control. A recurring theme was enhancing accessibility for elderly and disabled users, demonstrating the potential of gesture-based interfaces to create inclusive and intuitive smart environments.

Motivation

- Growing demand for **touchless technologies** due to hygiene concerns.
- Need for **inclusive solutions** for elderly or physically challenged individuals.
- Desire to develop a **low-cost and scalable system** that can be extended to more appliances.
- Interest in leveraging **computer vision libraries like Mediapipe** for practical IoT.

Objectives

1. To design a smart home automation system controlled via hand gestures.
2. To implement gesture recognition using the **Mediapipe library** in Python.
3. To use a Raspberry Pi 3B+ as the processing and control unit.
4. To integrate a **3-channel relay module** for controlling three light bulbs.

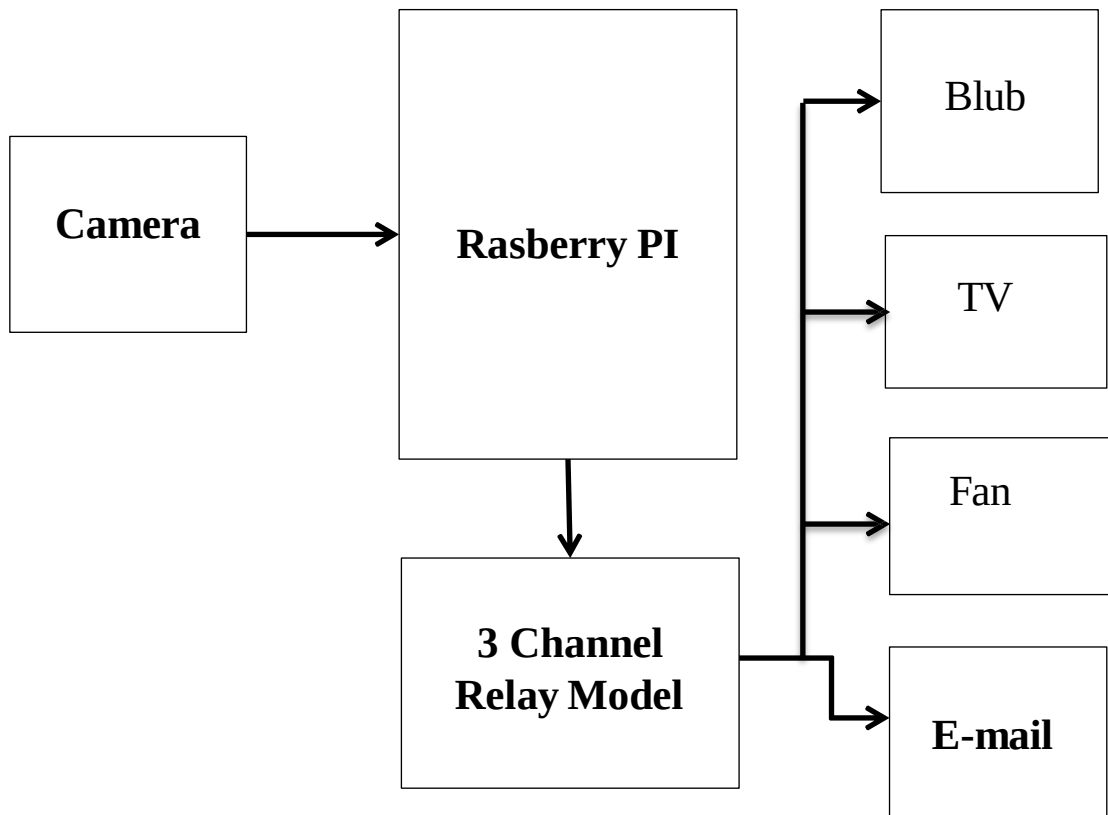
Problem Statement

Existing smart home automation systems rely on either physical touch, mobile apps, or voice-based control, which may not always be feasible. For instance:

- Voice assistants may fail in noisy environments.
- Smartphones may not always be within reach.
- Touch controls pose hygiene and accessibility challenges.

there is a need for a **gesture-based control mechanism** that is simple, cost-effective, and reliable for controlling household appliance

METHODOLOGY



BLOCK DIAGRAM

Block Diagram Description

The smart home automation system based on hand gesture recognition using feature fission and a recurrent neural network (RNN) operates through a sequence of well-defined stages. Initially, the system captures real-time hand gestures using an image or depth-sensing device such as a camera or Kinect sensor.

The captured data is then sent to a preprocessing module, where background noise is removed, the hand is segmented from the rest of the image, and the input is normalized to ensure consistency in size, orientation, and lighting conditions.

Once the data is preprocessed, it is passed to the feature extraction module, which uses a technique called feature fission to divide the input into distinct components — typically spatial features (such as hand shape and contours) and temporal features (such as motion trajectory and dynamics). These separated feature streams allow the system to emphasize and process different aspects of the gesture more effectively. By integrating these three modules, the proposed solution creates a holistic framework that addresses the challenges of monitoring vessel activities, detecting encroachments, and managing information related to violators within port limits. The amalgamation of precise navigation, advanced surveillance capabilities, and obstacle detection technologies offers a streamlined and technologically advanced solution, significantly enhancing the effectiveness of port management and security.

Proposed System

The proposed system uses a **Raspberry Pi 3B+** connected to a camera for capturing real-time hand gestures. Using the **Mediapipe library**, hand landmarks are detected and classified into specific gestures. Each gesture corresponds to an appliance control action (turning ON/OFF three light bulbs). A **3-channel relay module** is interfaced with the Raspberry Pi to switch the electrical appliances.

Expected Output

- The system should successfully detect hand gestures in real time.
- Recognized gestures should trigger corresponding relays to control three light bulbs
- The system should operate smoothly with minimal delay.

Component Requirements

Hardware Requirements

- Raspberry Pi 3B+
- Pi Camera / USB Camera
- 3-Channel Relay Module $3 \times$ Light Bulbs (220V)

- Jumper wires, Power supply

Software Requirements

- Raspbian OS
- Python 3.x
- Mediapipe Library
- OpenCV
- RPi.GPIO library

Conclusion

This project demonstrates a smart home automation system using hand gestures, leveraging computer vision and embedded systems. The use of the Mediapipe library for gesture detection ensures high accuracy and efficiency, while the Raspberry Pi 3B+ provides an affordable computing platform. The system effectively controls three appliances using intuitive hand gestures, eliminating the need for touch or voice. With further development, the system can be scaled to a full-fledged smart home ecosystem with IoT integration.

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